

Technology Stack

Date	26 June 2026
Team ID	XXXXXX
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	2 Marks

Technology Stack Details

S.No	Component	Technology	Purpose
1	Frontend	HTML5, CSS3, JavaScript, Jinja2	User Interface
2	Backend	FastAPI	REST APIs
3	Authentication	JWT	Secure Sessions
4	Database	SQLite	Data Storage
5	AI Models	Gemini 1.5 Pro	Question Answering
6	Local Models	LaMini-Flan-T5	Concept Explanation
7	API Layer	FastAPI Routers	Service Communication
8	Hosting	Render	Deployment
9	Version Control	Git & GitHub	Collaboration
10	Testing	Pytest	Module Testing

Third Party APIs

Google Gemini API

Generates:

- Educational Answers
- Concept Explanations
- Quiz Questions
- Summaries
- Learning Recommendations

Tech Stack Justification

Frontend → Interactive Student Dashboard

Backend → FastAPI for high-performance APIs

Database → SQLite for lightweight storage

AI → Gemini 1.5 Pro for educational intelligence

Deployment → Render

Version Control → GitHub

Testing → Pytest